Text Books

1. Nag, P.K., Power Plant Engineering, McGraw-Hill (2007) 3rd ed.
2. Raja, A.K., Srivastava, A.P. & Dwivedi, M., Power Plant Engineering, New Age Int. (2006) 1st ed.

Reference Books

1. Elanchezhian, C., Saravankumar, L., Ramnath, B. V., Power Plant Engineering, I-K Int. (2007) 1st ed.
2. Elliot, T.C., Chen, K., Swanekamp, R., Stanadard Handbook of Power Plant Engineering, McGrawhill Education (1998) 1st ed.

Evaluation Scheme:

S.No.	**Evaluation Elements**	**Weightage (%)**
1.	MST	20
2.	EST	40
3.	Sessionals	40

UME419 : **Renewable Energy Systems**

L	T	P	Cr
3	0	0	3.0

Course Objective: This course introduces various types of renewable energy resources, their characteristics and their advantages over conventional fuels. This course also introduces the technologies for harnessing these energy resources by using simple to advanced energy systems.

Syllabus

Introduction: Energy demand and availability, energy resources, environmental impact of conventional energy usage, heat and fluid flow concepts for energy systems.

Solar Energy: Introduction, extraterrestrial solar radiation, radiation at ground level, collectors- solar cells, applications of solar energy, types of solar collectors, storage and utilization, solar water heating systems, solar driers, solar thermal power systems, solar photovoltaics.

Energy from Biomass: Producer gas, bio-gas, bio-diesel and bio-ethanol.

Wind, Geo-thermal and Hydro Energy Sources: Wind energy systems, wind mill & farms, performance and economics, geothermal power plants, tidal power plants, Micro and small hydro energy systems.

Other Renewable Energy Resources: Thermoelectric conversion system, thermos-ionic conversion system, photo voltaic power system, fuel cells, magneto-hydrodynamic system, integrated energy systems, system design, economics of renewable energy systems.

Research Assignment: Students in a group will submit a research assignment on the following topics:

- Application of solar energy for industrial process heating, desalination and cooling.
- Innovative applications of renewable energy to reduce the consumption of conventional fuels.
- Performance and Emission Characteristics of a Diesel Engine fueled with bio-diesel, bio-gas and producer gas.

Research assignment will constitute collection of literature from library/internet, plant visit and formulation and analysis of the problem.

(10% weightage of total marks shall be given to this assignment).

Course Learning Objectives (CLO)

1. The students will be able to:
2. Calculate the terrestrial solar radiation on an arbitrary tilted surface.
3. Use flat plate solar collector mathematical model to calculate the efficiency and performance parameters of the same.
4. Determine the plant efficiency of geothermal power plant.
5. Select the factors that are required to consider when selecting sites for tapping renewable energy.
6. Determine maximum efficiency and maximum obtainable power from a given wind turbine

1.

Text Books

1. Duffie, J.A. and Beckmann, W.A., Solar Engineering of Thermal Processes, John Wiley (2006) 3rd ed.